BOAT AND STREAM

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BOAT AND STREAM

- Boat and Stream – Concept
- There are a variety of sub concepts that are related to answering questions based on boat and streams concept. Given below are the four terms which are important for a candidate to know to understand the concept of streams.
- **Stream –** The moving water in a river is called a stream.
- **Upstream –** If the boat is flowing in the opposite direction to the stream, it is called upstream. In this case, the net speed of the boat is called the upstream speed
- **Downstream –** If the boat is flowing along the direction of the stream, it is called downstream. In this case, the net speed of the boat is called downstream speed
- **Still Water –** Under this circumstance the water is considered to be stationary and the speed of the water is zero

BOAT AND STREAM

- Upstream and Downstream – Formula
- Given below are a few important formulas with the help of which you can solve the questions based on boat and streams.
- Candidates must learn these formulas by heart to ensure they are able to answer the simple formula based questions correctly and do not end up losing marks for direct questions.
- **Upstream = (u–v) km/hr**, where “u” is the speed of the boat in still water and “v” is the speed of the stream
- **Downstream = (u+v)Km/hr**, where “u” is the speed of the boat in still water and “v” is the speed of the stream
- **Speed of Boat in Still Water = ½ (Downstream Speed + Upstream Speed)**
- **Speed of Stream = ½ (Downstream Speed – Upstream Speed)**
- **Average Speed of Boat = {(Upstream Speed × Downstream Speed) / Boat’s Speed in Stii Water}**

BOAT AND STREAM

Q 1. The speed of a boat in still water is 2 km/hr. If its speed upstream be 1 km/hr., then speed of then stream is–

(1) 2 km/hr.

(2) 3 km/hr.

(3) 1 km/hr.

(4) 1.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 1. The speed of a boat in still water is 2 km/hr. If its speed upstream be 1 km/hr., then speed of then stream is–

(1) 2 km/hr.

(2) 3 km/hr.

(3) 1 km/hr.

(4) 1.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 2. A boat goes 14 km upstream in 56 minutes. The speed of stream is 2 km/hr. The speed of boat in still water is–

(1) 6 km/hr.

(2) 15 km/hr.

(3) 14 km/hr.

(4) 17 km/hr.

(5) None of these

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BOAT AND STREAM

Q 2. A boat goes 14 km upstream in 56 minutes. The speed of stream is 2 km/hr. The speed of boat in still water is–

(1) 6 km/hr.

(2) 15 km/hr.

(3) 14 km/hr.

(4) 17 km/hr.

(5) None of these

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BOAT AND STREAM

Q 3. The speed of a boat in still water is 10 km/hr. If its speed downstream be 13 km/hr., then speed of the stream is :

(1) 1.5 km/hr.

(2) 3 km/hr.

(3) 11.5 km/hr.

(4) 5.75 km/hr.

(5) None of these

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BOAT AND STREAM

Q 3. The speed of a boat in still water is 10 km/hr. If its speed downstream be 13 km/hr., then speed of the stream is :

(1) 1.5 km/hr.

(2) 3 km/hr.

(3) 11.5 km/hr.

(4) 5.75 km/hr.

(5) None of these

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BOAT AND STREAM

Q 4. The rowing speed of man in still water is 20 km/hr. going downstream, he moves at the rate of 25 km/hr. The rate of stream is–

(1) 45 km/hr.

(2) 2.5 km/hr.

(3) 12.5 km/hr.

(4) 5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 4. The rowing speed of man in still water is 20 km/hr. going downstream, he moves at the rate of 25 km/hr. The rate of stream is–

(1) 45 km/hr.

(2) 2.5 km/hr.

(3) 12.5 km/hr.

(4) 5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 5. A man can row downstream at the rate of 14 km/hr. and upstream at 5 km/hr. Find man's rate in still water.

(1) 9.5 km/hr.

(2) 8 km/hr.

(3) 8.5 km/hr.

(4) 9 km/hr.

(5) None of these

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BOAT AND STREAM

Q 5. A man can row downstream at the rate of 14 km/hr. and upstream at 5 km/hr. Find man's rate in still water.

(1) 9.5 km/hr.

(2) 8 km/hr.

(3) 8.5 km/hr.

(4) 9 km/hr.

(5) None of these

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BOAT AND STREAM

Q 6. A man can row downstream at the rate of 16 km/hr and upstream at 11 km/hr. Find man's rate in still water.

(1) 14 km/hr.

(2) 13.5 km/hr.

(3) 14.5 km/hr.

(4) 15.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 6. A man can row downstream at the rate of 16 km/hr and upstream at 11 km/hr. Find man's rate in still water.

(1) 14 km/hr.

(2) 13.5 km/hr.

(3) 14.5 km/hr.

(4) 15.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 7. A boat moves with a speed of 11 km per hour along the stream and 7 km per hour against the stream. The rate of the stream is _____ km/hr.

(1) 1 km/hr.

(2) 1.5 km/hr.

(3) 2 km/hr.

(4) 2.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 7. A boat moves with a speed of 11 km per hour along the stream and 7 km per hour against the stream. The rate of the stream is _____ km/hr.

(1) 1 km/hr.

(2) 1.5 km/hr.

(3) 2 km/hr.

(4) 2.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 8. A man rows upstream 11 km and downstream 26 km taking 5 hours each time. The velocity of the current is _____ km/hr.

(1) 1 km/hr.

(2) 1.3 km/hr.

(3) 1.5 km/hr.

(4) 2.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 8. A man rows upstream 11 km and downstream 26 km taking 5 hours each time. The velocity of the current is _____ km/hr.

(1) 1 km/hr.

(2) 1.3 km/hr.

(3) 1.5 km/hr.

(4) 2.5 km/hr.

(5) None of these

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BOAT AND STREAM

Q 9. A man can row 4.5 km/hr. in still water and he finds that it takes him twice as long to row up as to row down the river. find the rate of stream.

(1) 2 km/hr.

(2) 1.5 km/hr.

(3) 2.5 km/hr.

(4) 1.75 km/hr.

(5) None of these

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BOAT AND STREAM

Q 9. A man can row 4.5 km/hr. in still water and he finds that it takes him twice as long to row up as to row down the river. find the rate of stream.

(1) 2 km/hr.

(2) 1.5 km/hr.

(3) 2.5 km/hr.

(4) 1.75 km/hr.

(5) None of these

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BOAT AND STREAM

Q 10. A man can row 6 km/hr. in still water. It takes him twice as long to row up as to row down the river. Find the rate of the stream.

(1) 2 km/hr.

(2) 3 km/hr.

(3) 1.5 km/hr.

(4) 1 km/hr.

(5) None of these

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BOAT AND STREAM

Q 10. A man can row 6 km/hr. in still water. It takes him twice as long to row up as to row down the river. Find the rate of the stream.

(1) 2 km/hr.

(2) 3 km/hr.

(3) 1.5 km/hr.

(4) 1 km/hr.

(5) None of these

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BOAT AND STREAM

Q 11. The speed of a boat in still water is 15 km/hr. and the rate of current is 3 km/hr. The distance travelled downstream in 12 minutes is–

(1) 3.6 km

(2) 2.4 km

(3) 1.2 km

(4) 1.8 km

(5) None of these

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BOAT AND STREAM

Q 11. The speed of a boat in still water is 15 km/hr. and the rate of current is 3 km/hr. The distance travelled downstream in 12 minutes is–

(1) 3.6 km

(2) 2.4 km

(3) 1.2 km

(4) 1.8 km

(5) None of these

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BOAT AND STREAM

Q 12. Speed of a boat in standing water is 7 km/hr. and the speed of the stream is 1.5 km/hr. A distance of 7.7 km, going upstream is covered in–

(1) 1 hr. 15 min

(2) 1 hr. 12 min.

(3) 1 hr. 24 min

(4) 2 hr. 6 min

(1) (5) None of these

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BOAT AND STREAM

Q 12. Speed of a boat in standing water is 7 km/hr. and the speed of the stream is 1.5 km/hr. A distance of 7.7 km, going upstream is covered in–

(1) 1 hr. 15 min

(2) 1 hr. 12 min.

(3) 1 hr. 24 min

(4) 2 hr. 6 min

(1) (5) None of these

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BOAT AND STREAM

Q 13. A boat travels upstream from B to A and downstream from A to B in 3 hours. If the speed of the boat in still water is 9 km/hr. and the speed of the current is 3 km/ hr., the distance between A and B (in km) is–

(1) 4

(2) 6

(3) 8

(4) 12

(5) None of these

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BOAT AND STREAM

Q 13. A boat travels upstream from B to A and downstream from A to B in 3 hours. If the speed of the boat in still water is 9 km/hr. and the speed of the current is 3 km/ hr., the distance between A and B (in km) is–

(1) 4

(2) 6

(3) 8

(4) 12

(5) None of these

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BOAT AND STREAM

Q 14. A man can row 6 km/hr. in the still water. If the river is running at 2 km/hr., it takes him 3 hours to row to a place and back. How far is the place?

(1) 8 km

(2) 12 km

(3) 9 km

(4) 6 km

(5) None of these

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BOAT AND STREAM

Q 14. A man can row 6 km/hr. in the still water. If the river is running at 2 km/hr., it takes him 3 hours to row to a place and back. How far is the place?

(1) 8 km

(2) 12 km

(3) 9 km

(4) 6 km

(5) None of these

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BOAT AND STREAM

Q 15. The rate of flow of river water is 4 km/hr. A boat goes 6 km and back to the starting point in 2 hours. Find the speed of the boat in still water.

(1) 6 km/hr.

(2) 8 km/hr.

(3) 9 km/hr.

(4) 10 km/hr.

(5) None of these

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BOAT AND STREAM

Q 15. The rate of flow of river water is 4 km/hr. A boat goes 6 km and back to the starting point in 2 hours. Find the speed of the boat in still water.

(1) 6 km/hr.

(2) 8 km/hr.

(3) 9 km/hr.

(4) 10 km/hr.

(5) None of these

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BOAT AND STREAM

Q 16. In a stream running at 2 km/hr., a motorboat goes 12 km upstream and back again to the starting point in 2.5 hours. Find the speed of the motorboat in still water.

(1) 15 km/hr.

(2) 12 km/hr.

(3) 10 km/hr.

(4) 9 km/hr.

(5) None of these

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BOAT AND STREAM

Q 16. In a stream running at 2 km/hr., a motorboat goes 12 km upstream and back again to the starting point in 2.5 hours. Find the speed of the motorboat in still water.

(1) 15 km/hr.

(2) 12 km/hr.

(3) 10 km/hr.

(4) 9 km/hr.

(5) None of these

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BOAT AND STREAM

Q 17. A man can row 45 km upstream and 66 km downstream in 15 hrs. Also, he can row 65 km upstream and 77 km downstream in 20 hrs. Find the speed of the man in still water and rate of the current.

(1) 8 km/hr., 3 km/hr.

(2) 11 km/hr., 3 km/hr.

(3) 11 km/hr., 8 km/hr.

(4) 9 km/hr., 2 km/hr.

(5) None of these

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BOAT AND STREAM

Q 17. A man can row 45 km upstream and 66 km downstream in 15 hrs. Also, he can row 65 km upstream and 77 km downstream in 20 hrs. Find the speed of the man in still water and rate of the current.

(1) 8 km/hr., 3 km/hr.

(2) 11 km/hr., 3 km/hr.

(3) 11 km/hr., 8 km/hr.

(4) 9 km/hr., 2 km/hr.

(5) None of these

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BOAT AND STREAM

Q 18. Ajay can row a certain distance downstream in 5 hours and return the same distance in 7 hours. If the stream follows at the rate of 2 km per hour find the speed of Ajay in still water.

(1) 12 km/hr.

(2) 10 km/hr.

(3) 18 km/hr.

(4) 16 km/hr.

(5) None of these

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BOAT AND STREAM

Q 18. Ajay can row a certain distance downstream in 5 hours and return the same distance in 7 hours. If the stream follows at the rate of 2 km per hour find the speed of Ajay in still water.

(1) 12 km/hr.

(2) 10 km/hr.

(3) 18 km/hr.

(4) 16 km/hr.

(5) None of these

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BOAT AND STREAM

Q 19. A man row at a speed of 8 km/hr. in still water to a certain distance upstream and back to the starting point in a river which flows at 4 km/hr. Find his average speed for total journey.

(1) 8 km/hr.

(2) 6 km/hr.

(3) 4 km/hr.

(4) 10 km/hr.

(5) None of these

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BOAT AND STREAM

Q 19. A man row at a speed of 8 km/hr. in still water to a certain distance upstream and back to the starting point in a river which flows at 4 km/hr. Find his average speed for total journey.

(1) 8 km/hr.

(2) 6 km/hr.

(3) 4 km/hr.

(4) 10 km/hr.

(5) None of these

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BOAT AND STREAM

Q 20. A man can row 5 km/hr. in still water. If the river is running at 1 km/hr., it takes 2 hours more in upstream than to go downstream for the same distance. How far is the place?

(1) 24 km

(2) 20 km

(3) 18 km

(4) 16 km

(5) None of these

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BOAT AND STREAM

Q 20. A man can row 5 km/hr. in still water. If the river is running at 1 km/hr., it takes 2 hours more in upstream than to go downstream for the same distance. How far is the place?

(1) 24 km

(2) 20 km

(3) 18 km

(4) 16 km

(5) None of these

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BOAT AND STREAM

Q 21. A man can row 7 km/hr. in still water. If the river is running at 3 km/hr., it takes 6 hours more in upstream than to go downstream for the same distance. How far is the place?

(1) 48 km

(2) 36 km

(3) 42 km

(4) 40 km

(5) None of these

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BOAT AND STREAM

Q 21. A man can row 7 km/hr. in still water. If the river is running at 3 km/hr., it takes 6 hours more in upstream than to go downstream for the same distance. How far is the place?

(1) 48 km

(2) 36 km

(3) 42 km

(4) 40 km

(5) None of these

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THANK YOU

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